

CSCI 423 Fall 2017

HW1

Model Answer

$$\begin{aligned} \text{a. } \sum_{i=1}^N (3i + 7) &= 3 * \frac{N * (N + 1)}{2} + 7N \\ &= \frac{3N^2 + 17N}{2} \end{aligned}$$

$$\begin{aligned} \text{b. } \sum_{i=1}^N (i^2 - 2i) &= \frac{N * (N + 1) * (2N + 1)}{6} - 2 \frac{N * (N + 1)}{2} \\ &= \frac{2N^3 - 3N^2 - 5N}{6} \end{aligned}$$

$$\begin{aligned} \text{c. } \sum_{i=7}^N i &= \sum_{i=1}^N i - \sum_{i=1}^6 i \\ &= \frac{N * (N + 1)}{2} - 21 \end{aligned}$$

$$\begin{aligned} \text{d. } \sum_{i=5}^N (2i^2 + 1) &= 2 \sum_{i=1}^N i^2 + \sum_{i=1}^N 1 - 2 \sum_{i=1}^4 i^2 - \sum_{i=1}^4 1 \\ &= 2 \frac{N * (N + 1) * (2N + 1)}{6} + N - 30 - 4 \\ &= \frac{2N^3 + 3N^2 + 4N}{3} - 34 \end{aligned}$$

$$\begin{aligned} \text{e. } \sum_{i=1}^N 6^i &= \sum_{i=0}^N 6^i - 6^0 \\ &= \frac{6^{N+1} - 1}{5} - 1 \end{aligned}$$

$$\begin{aligned} \text{f. } \sum_{i=7}^N 4^i &= \sum_{i=0}^N 4^i - \sum_{i=0}^6 4^i \\ &= \frac{4^{N+1} - 1}{3} - 5461 \end{aligned}$$

one line.

$$n!$$

$$2^n, 2^{n-1}$$

$$(3/2)^n$$

$$n^3 + \lg n, n^3, n - n^2 + 5n^3$$

$$n^2$$

$$n \lg n$$

$$n$$

$$\sqrt{n}$$

$$(\lg n)^2$$

$$\lg n$$

$$\lg \lg n$$

$$6$$

2.

a. $g = O(f)$

b. $f = O(g)$

c. $f = O(g)$

d. $f = O(g)$

e. $f = O(g)$

f. $g = O(f)$

g. $g = O(f)$

h. $f = O(g)$